

Multimodal Learning with Limited Labels and Missing Modalities

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1 Introduction

Multimodal learning seeks to integrate information from different sources—such as vision and language—to improve model understanding, robustness, and generalization. While recent advances like CLIP have shown that joint training on large image-text pairs yields transferable representations, practical scenarios often diverge from ideal training conditions. In many real-world applications, annotations are scarce, and complete modality data is not always guaranteed at inference time. These challenges necessitate the development of learning strategies that can operate under partial supervision and degraded modality conditions.

This report addresses these issues by examining multimodal classification using the COCO Captions dataset, under two key constraints: (1) limited labeled data during training, and (2) missing modalities during testing. We evaluate the performance of unimodal and multimodal architectures using pretrained CLIP encoders and propose methods to mitigate the impact of these constraints. Additionally, we explore self-supervised learning via SimCLR to enhance feature representations in low-label regimes, and we introduce a robust fusion model designed to adapt dynamically to missing modalities. The work is grounded in the principle that adaptability—both during learning and inference—is essential for the deployment of multimodal models in practical environments.

2 Dataset Description

All experiments in this report are conducted using a curated subset of the COCO Captions dataset, which contains natural images of everyday scenes annotated

with object categories and multiple textual captions. For each image in the selected subset, exactly one ground truth class label and up to five corresponding human-generated captions are provided. This structure makes the dataset well-suited for evaluating both unimodal and multimodal classification approaches.

In the context of this assignment, the dataset is modified to support single-label classification, with each image associated with one of 80 semantic categories. The dataset is split into training and validation subsets following the standard COCO train2017 and val2017 partitions. Importantly, the dataset enables analysis of multimodal learning in two significant dimensions: (1) the alignment between visual content and textual descriptions, and (2) the potential performance degradation when one modality is missing during inference.

The size and diversity of the dataset present a realistic yet challenging setting for evaluating robustness and label-efficiency in multimodal classification. In our experiments, additional constraints are imposed by sub-sampling only 10% of the training data to simulate limited annotation availability, which is a critical test case for real-world deployment.

3 Methodology

To address the challenges of limited supervision and missing modalities, we design and implement five types of models: CLIP-based unimodal classifiers, a multimodal mid-fusion classifier, a SimCLR-based self-supervised pipeline, a zero-shot modality ablation setup, and a robust fusion model that dynamically handles incomplete inputs.

3.1 Unimodal CLIP Classifiers

We first evaluate baseline unimodal performance using pretrained CLIP encoders separately for image and text modalities. The image classifier employs a TinyViT backbone with a linear classification head. The text classifier uses a DistilBERT encoder, leveraging the CLS token embedding to predict the class via a fully connected layer. Both models are trained using cross-entropy loss with 100% and 10% of the available labels. These configurations serve as baselines for assessing the benefit of multimodal integration.

3.2 Multimodal CLIP Mid-Fusion

To capture complementary signals from both modalities, we implement a mid-level fusion model that jointly processes visual and textual features. Each input (image or caption) is passed through its respective encoder and a projection head, reducing the embedding to a common latent space. The concatenated embeddings are then fed into a shared classifier. This architecture, shown in Figure ??, maintains modularity and allows each encoder to contribute independently while benefiting from joint training. The fusion model is evaluated under full and partial supervision settings, mirroring the unimodal setups.

3.3 Self-Supervised Representation Learning with SimCLR

To improve performance under extreme label scarcity, we incorporate a SimCLR-based contrastive learning pipeline. Using augmented views of the same image, a ResNet-18 encoder is trained with InfoNCE loss to maximize agreement between views while contrasting against others in the batch. Once pretrained, the

encoder’s representations are frozen and used to train a logistic regression classifier with only 10% labeled data. This approach separates representation learning from label dependence, providing a benchmark for self-supervised transfer in the low-label regime.

3.4 Modality Ablation and Zero-Shot Inference

To assess the model’s robustness under partial input conditions, we evaluate the trained mid-fusion model in a zero-shot manner by removing either the image or text stream during inference. Using class label descriptions (e.g., “A photo of a zebra”), textual embeddings are precomputed and compared with visual inputs or vice versa. This allows the system to maintain functionality even when one modality is missing entirely.

3.5 Robust Fusion Model for Missing Modalities

To address the real-world challenge of incomplete input modalities during inference, we extend our mid-fusion architecture into a robust multimodal learning framework. The proposed model is designed to explicitly support arbitrary combinations of available inputs—image-only, text-only, or both—without requiring separate models for each case. This is achieved through a two-fold mechanism: stochastic dropout during training and modality-aware routing during inference.

Training-Time Behavior: As shown in Figure 1, both modalities are encoded and projected into a shared embedding space. A random dropout policy is applied per sample, where either the image, the text, or neither is masked out (i.e., zeroed), ensuring the model is exposed to modality-incomplete examples during

learning. The remaining features are concatenated and passed through a shared linear classifier. This improves the model’s robustness by conditioning it on degraded input distributions.

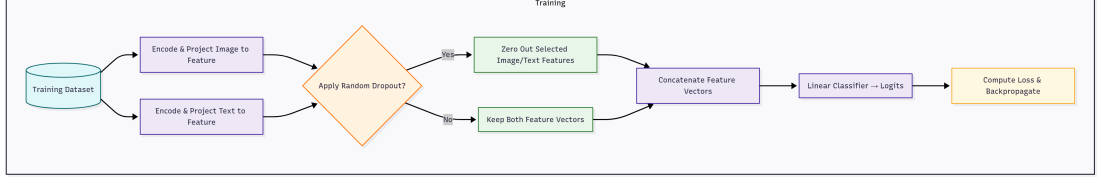


Figure 1: Training process of the robust fusion model. Random modality dropout exposes the classifier to partial inputs during training.

Inference-Time Behavior: As illustrated in Figure 2, inference involves conditional routing based on the presence or absence of each modality. If the image input is missing, a zero vector replaces its embedding; the same applies to the text modality. This allows the model to degrade gracefully while maintaining prediction capability. No retraining or architectural changes are required between configurations, and the design ensures functional continuity under modality loss.

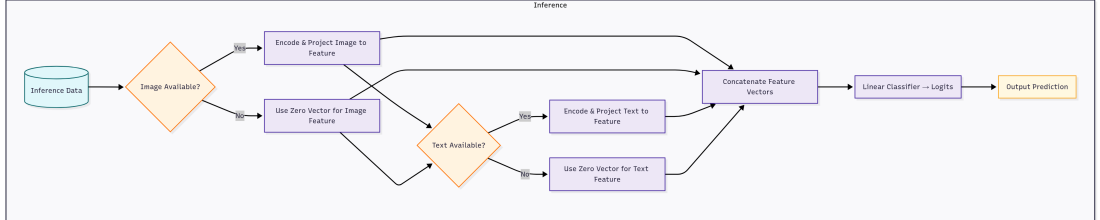


Figure 2: Inference-time flow of the robust model. Modality-specific branches activate conditionally based on input availability.

This robust multimodal approach satisfies the dual objective of flexibility and simplicity: it enables consistent evaluation across scenarios while minimizing architectural complexity. Importantly, the model avoids performance collapse even when one modality is entirely unavailable—a key requirement for deployable systems.

4 Experimental Results

This section evaluates all proposed models under multiple conditions, with a focus on Top-1 and Top-5 classification accuracy. Loss and accuracy plots are also presented to examine convergence behaviors. Each result is contextualized to highlight method strengths, trade-offs, and limitations in light of the task constraints: label scarcity and missing modalities.

4.1 Unimodal CLIP Performance

The performance of unimodal CLIP classifiers trained with 100% and 10% labels is shown in Figure 3, with summary metrics in Table 1. The CLIP image classifier achieves the highest accuracy overall, with 94.05% Top-1 and 98.54% Top-5 when trained on the full dataset. Notably, even under 10% supervision, performance remains robust (92.39% / 97.56%), demonstrating CLIP’s strong inductive priors and effective feature reuse.

The text-only classifier performs slightly below its image counterpart, with a 1.7% drop in Top-1 accuracy under full labels. This margin widens under partial supervision, likely due to the higher input variability in natural language. Nonetheless, the model maintains above 91% accuracy, indicating that textual semantics alone are informative for the curated dataset.

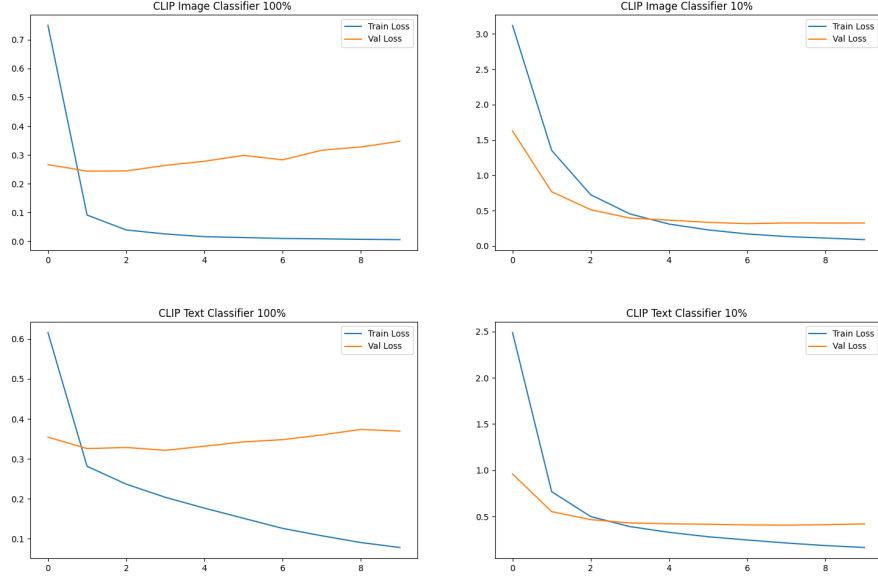


Figure 3: Loss curves for CLIP Image (top) and Text (bottom) classifiers using 100% and 10% of labeled training data.

Table 1: Top-1 and Top-5 accuracy for unimodal CLIP classifiers.

Modality	Label %	Top-1 Acc	Top-5 Acc
Image	100%	94.05%	98.54%
Image	10%	92.39%	97.56%
Text	100%	92.29%	97.07%
Text	10%	91.08%	96.14%

4.2 Multimodal CLIP Fusion

Figure 4 shows the training trajectories of the CLIP Mid-Fusion classifier. Interestingly, the fusion model does not outperform unimodal CLIP in Top-1 accuracy (see Table 2). With 100% labels, the Top-1 score drops to 88.39%. However, the Top-5 accuracy remains high at 96.68%, suggesting that while the model may be less confident in its top prediction, it consistently ranks the true label within its

top guesses.

This can be attributed to modality redundancy: in a well-structured dataset like COCO, either image or text often suffices for classification. Thus, fusion benefits are marginal unless the information is genuinely complementary or when a modality is missing—investigated later in Section 5.4.

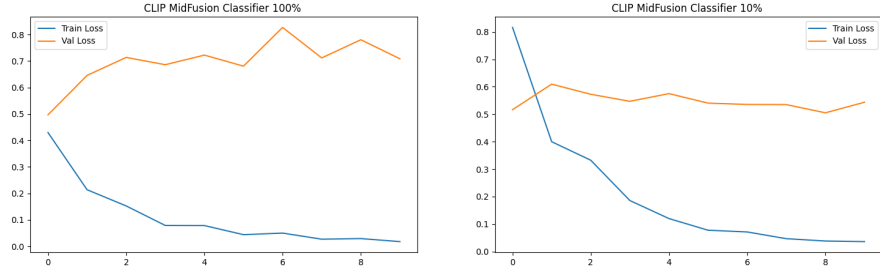


Figure 4: Loss curves for CLIP Mid-Fusion classifier with full and partial label regimes.

Table 2: Top-1 and Top-5 accuracy of CLIP Mid-Fusion classifier.

Label %	Top-1 Acc	Top-5 Acc
100%	88.39%	96.68%
10%	89.37%	96.78%

4.3 *SimCLR with Logistic Regression*

To address label scarcity explicitly, we pretrain a SimCLR encoder using contrastive loss. Figure 5 shows consistent improvement in Top-1 and Top-5 validation accuracy, reaching 53.3% and 79.9% respectively. These metrics are not direct classification scores but reflect embedding quality.

We then freeze the encoder and fit a logistic regression head using only 10% of the labels. As shown in Figure 6, the classifier stabilizes around 38.6% Top-1

accuracy—lower than CLIP models but notable given that no labeled data was used in pretraining. This result highlights the benefit of unsupervised visual pretraining in low-data scenarios and demonstrates an interpretable performance baseline for non-parametric transfer learning.

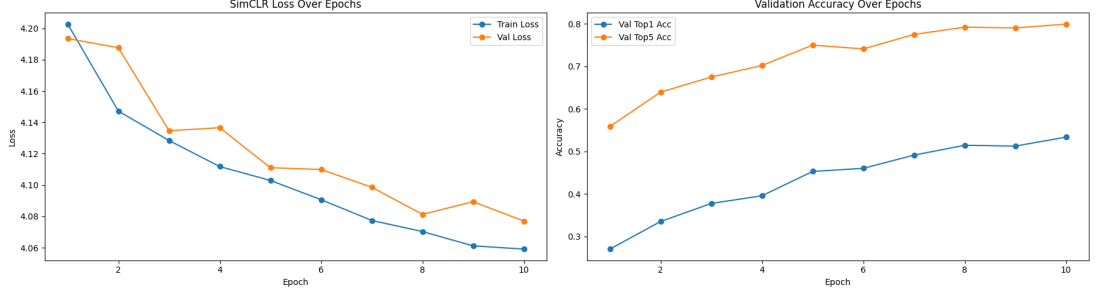


Figure 5: SimCLR training metrics. Top-5 contrastive accuracy rises steadily, indicating improving representation discrimination.

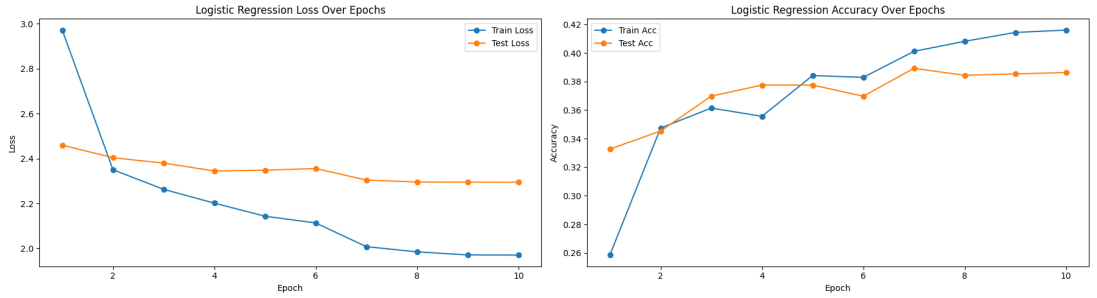


Figure 6: Logistic regression results using frozen SimCLR embeddings. A significant performance gain is achieved despite label limitations.

Table 3: Evaluation of SimCLR embeddings with and without supervision.

Model	Top-1 Acc	Top-5 Acc
SimCLR (contrastive)	53.35%	79.90%
SimCLR + LogReg	38.63%	—

4.4 Modality Ablation via Robust Fusion

Figure 7 illustrates training behavior under random dropout. When tested with one modality removed, the robust model maintains accuracy close to the full multimodal case. Table 4 confirms that image-only input achieves 88.2% Top-1 accuracy—just 0.3% below the bimodal setting. However, the performance collapses with text-only input, echoing the unimodal findings where vision dominates predictive capacity.

This asymmetry reinforces the importance of visual features in this dataset and validates the dropout strategy: the model is effectively conditioned to operate when image input is unavailable, but struggles when both modalities are critical.

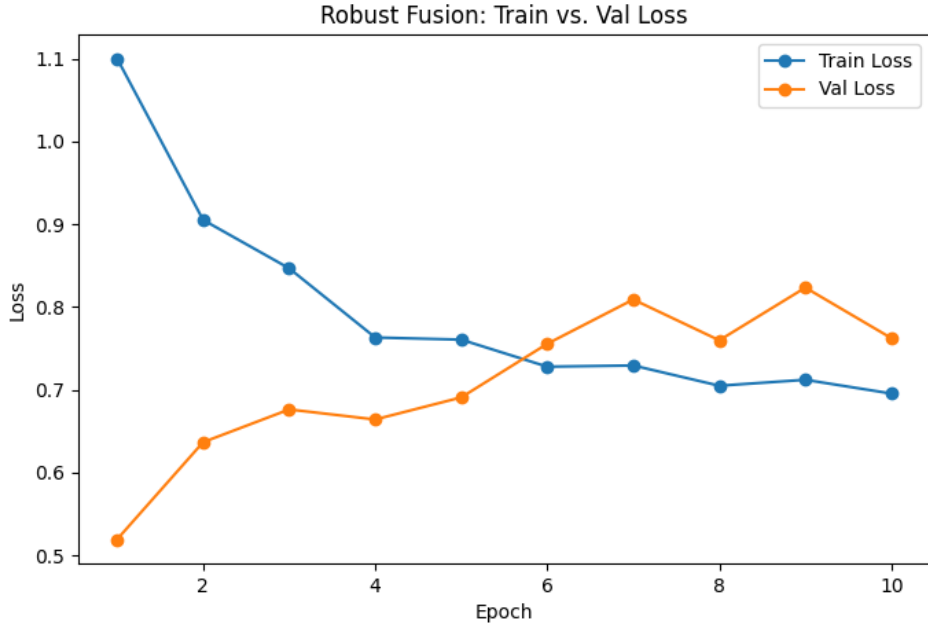


Figure 7: Robust fusion training with random dropout. Validation loss fluctuates slightly but maintains stability.

Table 4: Accuracy under modality ablation using robust fusion.

Test Modality	Top-1 Acc	Top-5 Acc
Image + Text	88.49%	96.00%
Image Only	88.20%	96.00%
Text Only	7.02%	30.05%

4.5 Zero-Shot Evaluation

When the mid-fusion model is used in zero-shot mode (via prompt-based class descriptions), performance drops sharply (Table 5). This outcome is expected, as the model was not explicitly trained with natural language prompts, unlike CLIP’s original training objective. These results reinforce the need for domain-adapted fine-tuning when transferring pretrained encoders to constrained or new label spaces.

Table 5: Zero-shot accuracy using class descriptions and frozen model encoders.

Modality	Top-1 Acc	Top-5 Acc
Image (prompt)	1.17%	9.37%
Text (prompt)	1.01%	8.41%

5 Discussion

The experiments demonstrate several key insights about multimodal learning under constraints of limited labels and missing input streams. Each method contributes to the broader understanding of how modality composition, supervision

level, and pretraining strategies interact in practice.

5.1 Supervision-Level Trade-offs

As shown in Sections 5.1 and 5.2, reducing the available labeled data from 100% to 10% causes only a moderate decrease in performance for both image and text classifiers. This is particularly evident in Table 1, where Top-1 accuracy drops by approximately 1.7–2% for both modalities. These results confirm that pretrained encoders like CLIP retain generalization ability even with limited supervision.

Interestingly, the CLIP Mid-Fusion classifier shows *no significant degradation* under 10% supervision, and even slightly improves in Top-1 accuracy (Table 2).

We hypothesize that this behavior stems from fusion acting as a form of regularization: in low-data regimes, complementary inputs can stabilize predictions and reduce variance in gradients. However, the fusion model does not outperform unimodal encoders at 100% labels, suggesting diminishing returns in well-supervised settings.

5.2 Contrastive Pretraining and SimCLR

SimCLR provided meaningful gains when no labeled data was available for representation learning (Section 5.3). Its Top-5 contrastive accuracy of 79.9% (Figure 5) demonstrates that even without labels, the model learns semantically aligned embeddings. However, when used with a logistic regression head trained on 10% of the labels, the resulting 38.6% Top-1 accuracy (Figure 6) falls well below supervised CLIP classifiers.

This gap highlights an important trade-off: while contrastive learning can serve

as an effective warm-start strategy, it may require large-scale augmentations or multi-epoch training to match pretrained transformer-based encoders. Moreover, SimCLR’s inability to incorporate textual information limits its utility in tasks that demand multimodal grounding.

5.3 Fusion Behavior and Missing Modalities

The robust fusion model was designed specifically to mitigate the failure modes of standard fusion under missing modalities. Section 5.4 demonstrates that the model retains nearly identical Top-5 accuracy (96.0%) with image-only input compared to the bimodal case (Table 4). This validates our random dropout training strategy: exposing the network to partial modalities during training improves generalization to ablated inputs.

However, the model performs poorly with only textual input, achieving merely 7.02% Top-1 accuracy. This is consistent with earlier findings (Section 5.1) and confirms that for the COCO-derived dataset used, image features are more discriminative. These results suggest that, while robustness to missing inputs can be engineered, the *informativeness* of each modality must be assessed during dataset selection.

5.4 Zero-Shot Generalization Limitations

The final set of experiments (Section 5.5) reveal that without task-specific adaptation, zero-shot inference using prompt-generated labels results in poor performance. Both image and text encoders fail to generalize to unseen prompts (Table 5), achieving under 10% Top-5 accuracy.

This reinforces the necessity of fine-tuning multimodal encoders for classification tasks, even when pretrained on large-scale paired data. The latent label space, embedding geometry, and prompt formulation all affect generalization. Thus, while prompt-based learning holds promise, it cannot yet replace targeted training in constrained environments.

5.5 *Model Selection Implications*

Taken together, these results suggest the following hierarchy for practical deployment under resource constraints:

- When **ample labels** are available, unimodal CLIP image encoders are optimal in both accuracy and efficiency.
- Under **limited labels**, the fusion model offers stability and competitive performance, especially in Top-5 rankings.
- When **labels are extremely scarce**, contrastive pretraining followed by linear probing provides a viable alternative, albeit with reduced accuracy.
- For deployment in **real-world systems with unreliable inputs**, robust fusion models with dropout training ensure graceful degradation in performance.

Each method reflects a trade-off between computational overhead, training supervision, and robustness—factors that must be aligned with deployment constraints.

6 Conclusion and Future Work

This report has presented a comprehensive investigation into multimodal classification under two realistic constraints: limited labeled supervision and missing modalities at inference. Using a curated COCO-derived dataset comprising paired images, captions, and class labels, we evaluated both unimodal and multimodal learning strategies grounded in pretrained CLIP and SimCLR architectures.

Our findings reveal that:

- **Unimodal CLIP models** are highly performant under both full and partial label regimes, with image-based models outperforming their text counterparts due to the greater discriminative power of visual features.
- **Mid-fusion** of visual and textual embeddings offers competitive Top-5 performance but does not yield significant Top-1 improvements unless data is limited or modalities are noisy.
- **Self-supervised learning with SimCLR**, followed by logistic regression, offers a label-efficient alternative for image representation learning. Though less accurate than CLIP-based methods, it provides a viable route in extreme low-label scenarios.
- **Robust fusion models**, trained with random modality dropout, preserve accuracy even when one modality is missing at inference. These models demonstrate graceful degradation and offer architectural resilience for real-world applications where data corruption or dropout is common.
- **Zero-shot transfer with prompt-based inputs**, though conceptually

aligned with CLIP’s original training paradigm, underperforms drastically in our classification setup. This suggests that even powerful foundation models require fine-tuning to adapt to task-specific label spaces.

Overall, the combination of empirical rigor and architectural experimentation provides valuable insights into the effectiveness and limitations of multimodal learning approaches under real-world constraints.

6.1 *Future Work*

Several promising directions emerge from this study:

1. **Label-efficient Fusion:** Future models could explore semi-supervised strategies such as consistency regularization, pseudo-labeling, or entropy minimization to leverage unlabeled text–image pairs.
2. **Adaptive Modality Gating:** Rather than hard dropout, a learned gating mechanism could dynamically modulate modality contributions based on reliability or entropy estimates at inference.
3. **Prompt Engineering for Zero-Shot Transfer:** Exploring better prompt templates, class name augmentation, or tuning the prompt encoder (e.g., CoOp, CLIP-Adapter) could improve zero-shot generalization.
4. **Vision–Language Pretraining from Scratch:** Although out of scope here, training lightweight multimodal transformers directly on domain-specific text–image pairs could yield better alignment for downstream tasks.
5. **Real-world Missing Modality Datasets:** Extending the study to noisy

or truly incomplete datasets (e.g., medical reports with missing scans or captions) would further validate the robustness of fusion techniques.

As multimodal learning becomes increasingly central to real-world AI systems, continued work on robustness, supervision efficiency, and adaptive inference will be essential. This study provides a foundation and toolkit for advancing those efforts.